

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech. Branch : Mechanical Engineering / Mechanical Engineering (Sandwich)

Subject Code & Name: BTMEC702 CAD/CAM

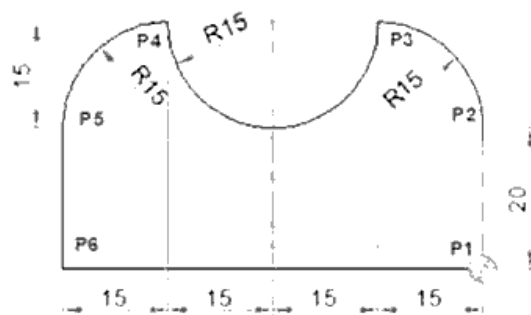
Semester : VII

Max Marks: 60

Date: 03/07/2024

Duration: 3 Hr.

	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		(Level/CO)
		Marks
Q. 1	Solve Any Two of the following.	12
A)	Explain minimum three Output devices used as CAD input device.	6
B)	State the various transformation matrices used for 3D transformations using homogeneous transformation system for A. Translation B. Rotation about X Axis C. Rotation about Y Axis D. Rotation about Z Axis E. Scaling F. Reflection about XY plane	6
C)	Describe any three methods for defining 'Circle' in a graphics software.	6
Q.2	Solve Any Two of the following.	12
A)	Explain B-Rep methods as a modeling technique for CAD.	6
B)	What are the various windowing applications in CAD? explain with the help of neat sketches and suitable examples.	6
C)	Explain various types of Robotic Configurations based on mobility.	6
Q. 3	Solve Any Two of the following.	12
A)	Explain "Straight line" control system used in CNC system. Also mention in which particular applications it will be recommended?	6
B)	Write NC program for finishing operation of turning of the job as shown in following figure. All dimensions are in mm	6
C)	Write NC program for finishing operation of milling of outer surface of the job as shown in following figure.	6



All the dimensions are in mm

Q.4	Solve Any Two of the following.		12
A)	Explain various steps in solving problem using Finite Element Analysis		6
B)	Explain the concept of 'Shape function'.		6
C)	Explain various types of 2D elements used in Finite Element Analysis		6
Q. 5	Solve Any Two of the following.		12
A)	What are the various essential components of a FMS system.		6
B)	What is Group Technology? What are its advantages.		6
C)	Differentiate the generative and retrieval type of CAPP system.		6
	*** End ***		